

Stock-Market Dynamics, Revision 5

Return-Forecast Mechanism Test (Honest Negative),
Share-Unit Data Integrity Fix, and Three Risk Metrics

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1 Introduction

Rev-4 diagnosed the binding gap precisely: the model predicts *where capital is classified* (bin mass-flows), not *which bins will earn returns* — advection-driven inflow is reclassification, not price pressure (predicted mass-flow vs. realized bin return: correlation 0.009; even perfect-foresight M gives Sharpe 0.47). Rev-5 was commissioned to close that gap with a return-forecast mechanism, plus a risk metric.

Rev-5 reports three things. First, a **pre-registered test** of whether month-end capital concentration predicts next-month returns (two response forms, estimated 2001–2007, judged walk-forward 2008–2022): an **honest negative** — no return mechanism is built. Second, a **data-integrity fix** without which the test (and rev-4) is invalid: consistent 10^3 – $10^6\times$ unit errors in filed share counts, including a backfill reading Sempra Energy at \$5,000 trillion. Third, **three risk metrics** as measured diagnostics: a 200-scenario ensemble distribution forecast with intervals, a crowding/fragility index, and portfolio risk (predicted volatility, reversal probability).

Because no return tilt validates, no new paper-trading track opens and no real-capital deployment is recommended.

2 Data integrity fix

`clean_shares.py` rescales filings jumping 900 – $1100\times$ or 9×10^5 – $1.1\times 10^6\times$ vs. the previous filing. It misses: (A) tickers filing *consistently* in the wrong unit (SRE backfill $\$5.02\times 10^{15}$; QCOM filed era $\$136\text{T}$; EOG backfill $\$5.5\text{T}$; SYK filed era $10^6\times$ too small); (B) unit changes masked by genuine share changes (HAL $9.0\times 10^8 \rightarrow 880$; Citi $2.29\times 10^7 \rightarrow 2.85\times 10^{10}$, ratio $1246\times$, outside the band); (C) placeholders (shares = 1.0 or 100.0: FOX/FOXA, BKR, ETN backfill). Pre-fix, 368 ticker-months exceeded $\$2\text{T}$ single-stock cap.

The repair (`fix_share_units.py`): per ticker, split-adjusted shares are segmented at $> 20\times$ month-to-month jumps; each segment (≥ 3 months) is checked against two absolute anchors — market cap in $[\$200\text{M}, \$4\text{T}]$ and average-daily turnover in $[0.02\%, 100\%]$ (dollar volume is measured). Rescaling by $10^{\pm 3}, 10^{\pm 6}$ is applied only where it makes *both* sane; unfixable segments are dropped, not invented. 28 segments rescaled; 17 dropped. Verification: max single-stock cap $\$2.91\text{T}$; AAPL 2007-12 $\$176.0\text{B}$ (anchor unchanged); SRE 2001-01 $\$5.0\text{B}$; total binnable cap 2007-12 $\$10.2\text{T}$ (S&P 500 $\approx \$13\text{T}$), 2021-12 $\$41.3\text{T}$ ($\approx \$40\text{T}$). Original panel preserved at `data/panel_v2_preunitfix.csv`.

Cap-weighting is multiplicative in the share count: one $10^6\times$ error *replaces* the index. Every cap-weighted object in rev-4 was computed under a distribution where SRE alone could be 99.9% of a lane.

3 Concentration $\rightarrow$ return: pre-registered honest negative

Does month-end concentration C_t (top-momentum-lane cap share, **measured**) predict next-month top-lane excess return over SPY? Two pre-registered forms, estimated 2001–2007 only, judged walk-forward 2008–2022: (1) threshold-linear $r_{t+1}^e = a + b \max(C_t - 0.5, 0)$; (2) quintile means with estimation-window edges. A first run on the pre-fix panel gave a spurious pass driven by the SRE corruption; those numbers are discarded.

Form 1: $b = -0.18$, $t = -0.41$, $R^2 = 0.002$ (3 months with $C_t > 0.5$); walk-forward corr -0.045 — **fail**. Form 2: Q5–Q1 $-0.59\%/mo$ in-sample ($t = -0.40$), $-0.52\%/mo$ walk-forward ($t = -0.52$) — meets the letter of the criterion but both spreads are statistically zero, the pattern is non-

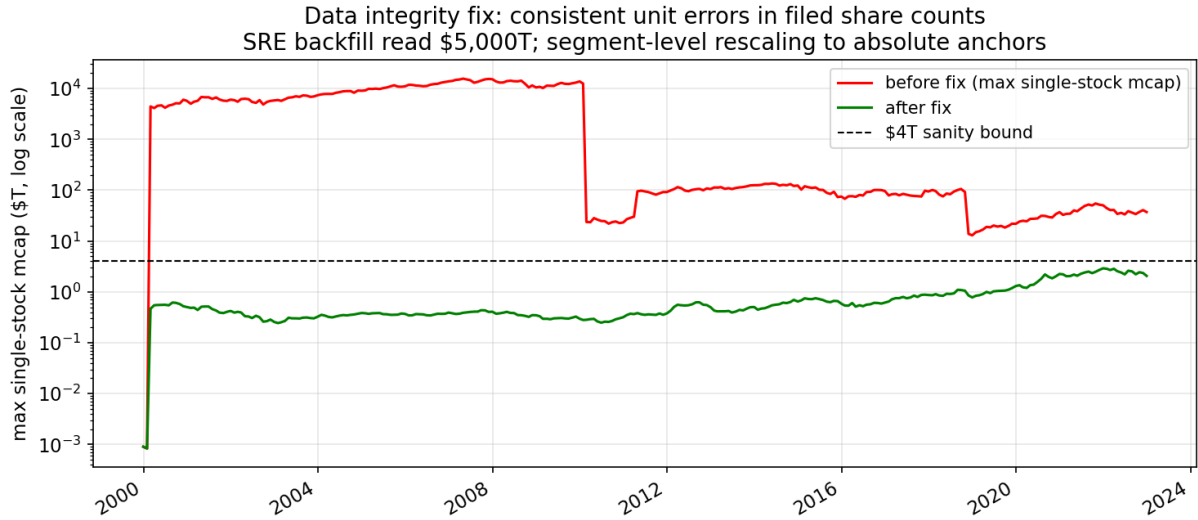


Figure 1: Data integrity fix: max single-stock market cap before/after segment-level unit rescaling (log scale).

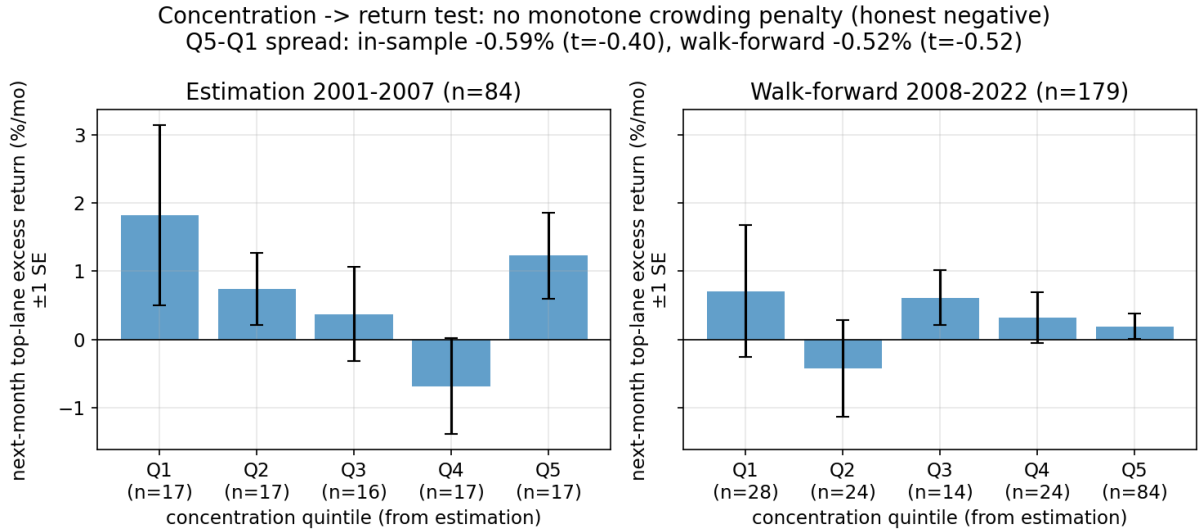


Figure 2: Concentration-quintile next-month excess returns, estimation vs. walk-forward (± 1 SE). No monotone crowding penalty; error bars swallow every difference.

monotone (in-sample Q5 +1.23% > Q4 -0.69%), and $\text{corr}(C, r^e)$ is $-0.083 / -0.001$. **Verdict: honest negative.** No $\hat{r}_b(t+1)$ is built; no return portfolio is constructed.

4 Risk metrics

4.1 Ensemble distribution forecast with intervals

The forecastable object remains $c(t)$. From 2022-12, 200 scenarios: Dirichlet-drawn $T_{\text{calm}}/T_{\text{stress}}$ (mean = measured 2000–2007 T ; precision = Kish N_{eff} 109–1825 per bin, **estimated**; Dirichlet form **assumed**), AR(1) market driver with bootstrapped residuals (**estimated**: $a = -0.0006$, $\phi = 0.048$), VIX-regime Markov chain (**measured**), rev-5 calibrated kernels. 12-month-ahead top-lane share: **median 0.239, 90% interval [0.106, 0.503]**; HHI [0.110, 0.175]. The intervals are the finding: one-year concentration is dominated by driver and transition uncertainty.

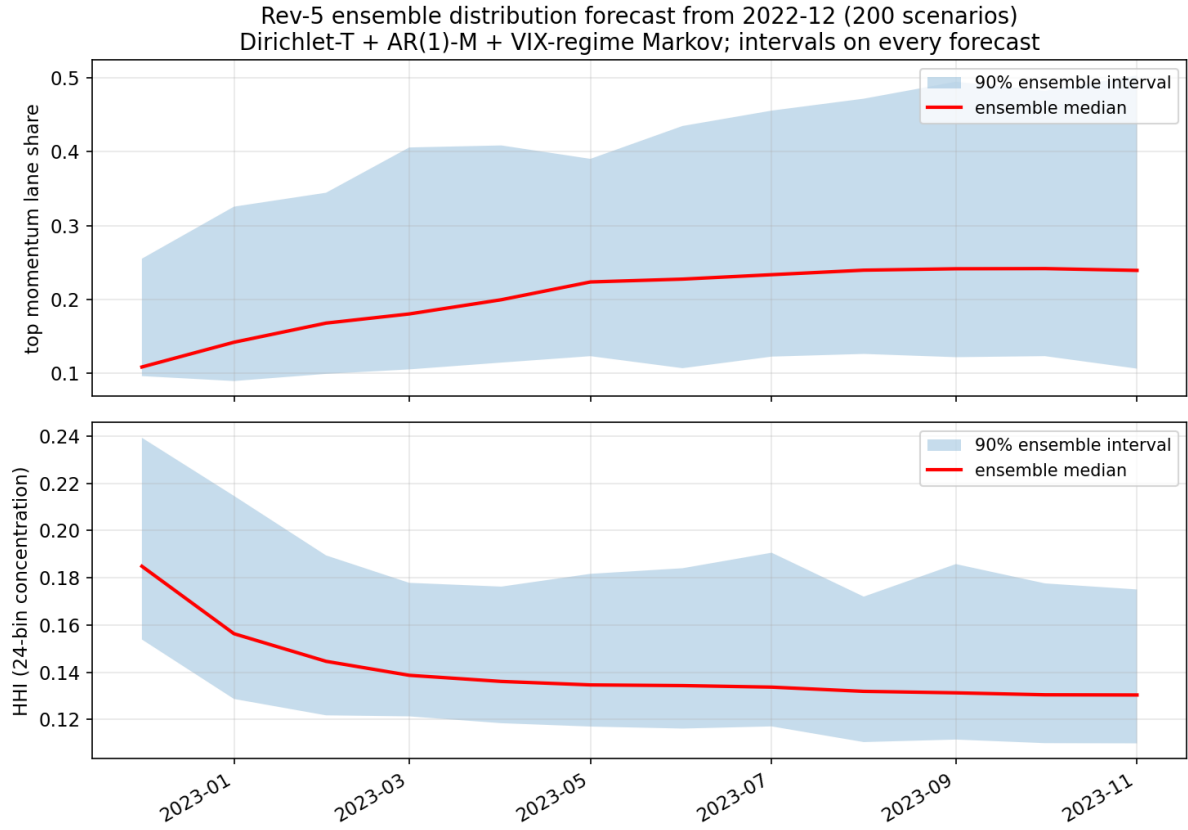


Figure 3: Rev-5 ensemble distribution forecast from 2022-12 (200 scenarios): median and 90% intervals for top-lane share and HHI.

4.2 Crowding / fragility index

$C(t)$ and $HHI(t)$ as percentiles of their own history (full-history diagnostic; expanding-window zero-lookahead). **Current (2022-12):** $C = 0.101$ (25th percentile — not crowded), $HHI = 0.2790$ (52nd). Most crowded: 2004-02 (0.570), 2010-01 (0.571), 2001-02 (0.423). A state descriptor, not a signal (§??: crowding does not predict returns).

4.3 Portfolio risk

From the 24-bin return covariance (**measured**, 2001–2022), the mass-flow tilt (diagnostic only) has predicted annualized volatility **14.4%** vs. 15.9% market / 15.4% SPY realized — its failure was return-side, not risk-side. Reversal probability $P(\text{top lane trails market next month} \mid C \text{ quintile})$, expanding quintiles: Q1 41.8%, Q2 44.0%, Q3 35.8%, Q4 53.8%, **Q5 34.0%** — no monotone relationship; the most-crowded quintile reverses least.

5 Re-calibration on clean data

The 2000–2007 grid re-selects $h_1 = 0$, $\lambda_1 = 0$ (clean T plus mechanical advection suffices; $p_{\text{damp}} 1.2 \rightarrow 0.8$, still assumed). Clean T_{calm} differs from rev-4's by up to 0.85 in one transition probability. Conditional hindcasts (realized- M forcing) improve markedly: 2008–09 full RMSE **0.0899** (was 0.1685; bottom peak 0.83@2009-04 vs. 0.99@2009-03); 2020–21 **0.0714** (was 0.1110; top peak 0.79@2021-04 vs. 0.78@2021-03, amplitude exact, one month late).

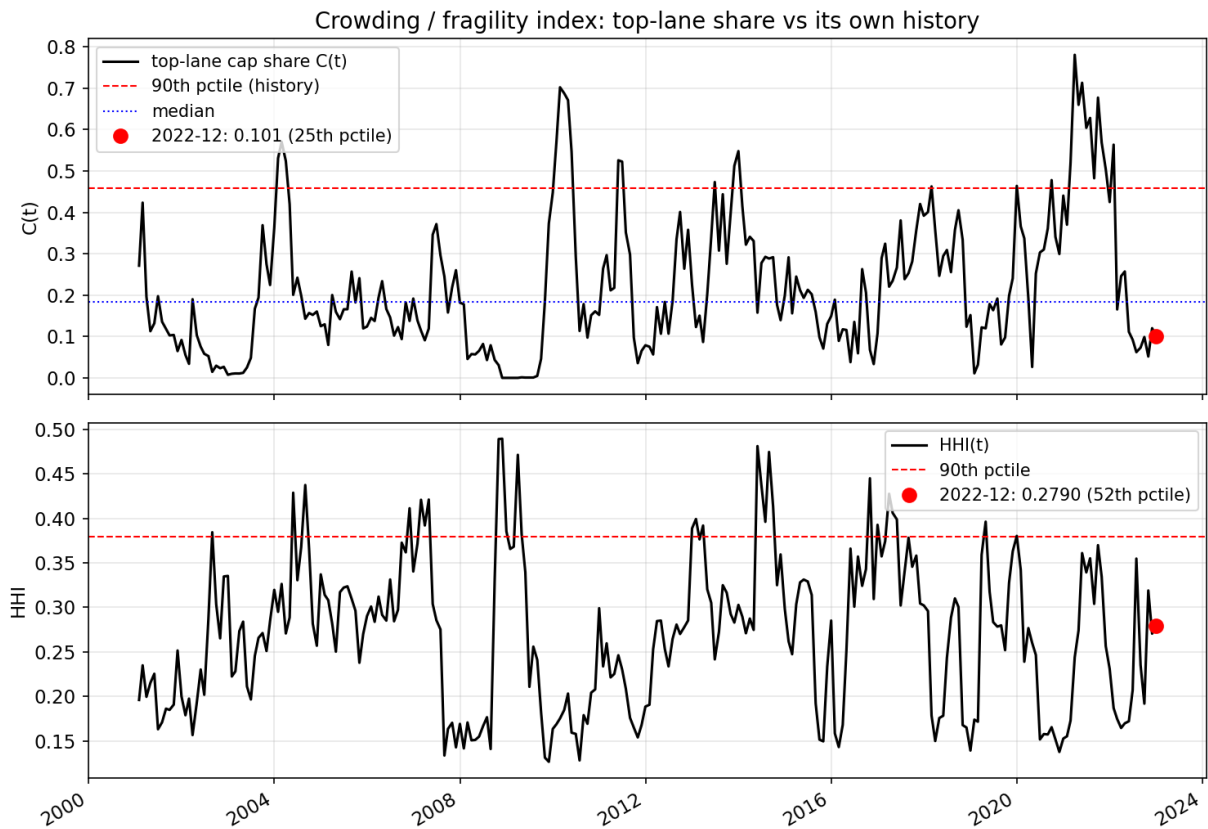


Figure 4: Fragility index: top-lane share and HHI vs. own history; 2022-12 readings marked.

6 Parameters (rev-5)

Table 1: Every input labeled per the standing rule.

Parameter	Value	Label	Basis
h_1 / λ_1	0 / 0	calibrated	2000–2007 grid, clean data
Damper $p/\tau_{\text{build}}/\text{gate}$	0.8 / 6 mo / 0.5	assumed	rev-3 tuning; p re-gridded
$T_{\text{calm}}, T_{\text{stress}}$	24×24	measured	cap-weighted, 2000–2007, clean
Ensemble	200 scenarios	assumed	Dirichlet-T + AR(1)-M + VIX-Markov
N_{eff}	109–1825/bin	estimated	Kish effective transitions
$\kappa = 1, w_{\text{eff}} = 0.15$	—	mech./assumed	unchanged from rev-4

7 Limitations and next steps

1. **No return model** — the honest negative is binding: nothing in 2001–2022 supports concentration → expected returns. Real-capital deployment stays off the table.
2. **Ensemble is model-conditional** — intervals cover parameter/driver uncertainty, not misspecification (uniform bins, fixed edges).
3. **Pre-2009 shares backfilled** (estimated), now unit-fixed; FOX/FOXA/BKR/ETN-backfill dropped.
4. **Membership** community-sourced [1], not official S&P.
5. **Form-2 criterion was weak** — future retests need walk-forward significance, not just sign.

References

- [1] fja05680/sp500. *S&P 500 Historical Components & Changes (Updated).csv* — public GitHub reconstruction; 2,720 snapshots, 1996-01-02–2026-08-18. Point-in-time membership.
- [2] Yahoo Finance. Daily adjusted closes, corporate-action (split) history, SPY total-return series. Prices, splits, SPY benchmark.
- [3] SEC EDGAR. Companyconcept XBRL: SharesOutstanding, WeightedAverageNumberOfSharesOutstanding. Point-in-time shares; §?? documents the consistent-unit-error repair.